FINAL REVISED RESEARCH BLUEPRINT

Hermitian dual-containing constacyclic codes over a commutative Frobenius ring and their 4-ary quantum-code images

Revised title	Hermitian Dual-Containing Constacyclic Codes over $F_{16}[U,V]/\langle U^k, V^4\rangle$ and 4-ary Quantum Codes
Key algebraic decision	Replace the matrix-ring alphabet with the distinct commutative ring $R_k = F_{16}[U,V]/\langle U^k, V^4\rangle$; do not assert an isomorphism.
Status	Blueprint proposal only. Expected theorems remain unproved; no code search or parameter catalogue is included.
Date	12 November 2026

Final scope: this is a substantially revised research blueprint. The original matrix-ring claims are not retained. The same cardinality is only a size match. Actual research execution is deferred until explicit approval.

Novelty status: a targeted search did not identify an exact match for the proposed ring-specific theorem package, but adjacent constacyclic/quantum literature is extensive. Novelty is conditional on a systematic review and completed proofs.

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Status: reconstructed plan; not a paper, not a theorem-proof dossier, and not yet an executed research project.
Date: 12 November 2026.
Core design decision: abandon the false identification with a matrix ring. Work instead over the explicitly defined commutative Frobenius ring below. It has the desired field and nilpotent coordinates and the same cardinality as the proposed matrix alphabet, but it is a different ring.

1. Final research title

Hermitian Dual-Containing Constacyclic Codes over $F_{16}[U,V]/\langle U^k,V^4\rangle$ and 4-ary Quantum Codes

The title deliberately does not say “over M_4.” No isomorphism between this commutative alphabet and $M_4(F_2[u]/\langle u^k\rangle)$ is claimed.

2. Research abstract

Let $R_k = F_{16}[U,V]/\langle U^k,V^4\rangle$, where $k\geq 2$. This is a finite commutative local Frobenius ring with residue field F_{16}, socle generated by $U^{k-1}V^3$, and 16^{4k} elements. It is not the matrix ring $M_4(F_2[u]/\langle u^k\rangle)$, despite the equality of their cardinalities.

The proposed project studies length-n λ-constacyclic ideals over R_k, for odd n and $\lambda\in F_{16}$ with $\lambda^5=1$. The order-five multipliers are central units in this commutative ring, and the Hermitian involution $a \mapsto a^4$ on F_{16}, fixing U,V, satisfies $\bar{\lambda}^{-1}=\lambda$. Since n is odd in characteristic two, $X^n-\lambda$ is square-free over F_{16}; its factorization gives an exact CRT decomposition into rings $K_i[U,V]/\langle U^k,V^4\rangle$, where K_i/F_{16} are finite field extensions.

The main structural target is a precise dual-containment criterion for a tractable but explicitly limited class: codes whose CRT components are monomial ideals. A Frobenius-functional construction supplies a duality-adapted F_{16}-linear coordinate map from R_k^n to F_{16}^{4kn}. Its intended properties are to preserve the Hermitian orthogonality relation exactly and to intertwine the ring constacyclic shift with a block constacyclic shift. A Hermitian dual-containing image then yields a 4-ary stabilizer code with dimension 2K-4kn; its distance is the minimum Hamming weight in the nontrivial logical cosets and must be certified separately.

Prior work already establishes broad constacyclic-to-quantum methods over chain and non-chain rings, and direct cyclic-code work over the original matrix-ring family. The proposed contribution is therefore not “the first quantum codes from constacyclic rings” or “the first Gray map preserving duality.” Its candidate novelty is the ring-specific CRT-orbit/monomial-annihilator criterion and explicit duality-adapted image for this two-nilpotent-variable complete intersection, followed—only if proofs and a systematic prior-art check support it—by verified quantum-code families. A targeted search did not surface an exact match for this ring and theorem package, but novelty remains conditional and uncertified.

3. Background and motivation

Finite-ring codes can expose algebraic structure that is difficult to see in an unstructured field-code search. Constacyclic invariance offers a polynomial quotient and, when the defining polynomial is separable, a component decomposition. Hermitian dual containment over F_{16}/F_4 is a standard route to 4-ary quantum stabilizer codes.

The original motivation combined a matrix alphabet with a proposed commutative local model. That identification fails. The revised project preserves the intended algebraic-code-to-quantum-code connection, the F_{16} Hermitian structure, the nilpotent directions, order-five multipliers, and the target F_4 quantum alphabet, while treating the commutative ring as a new alphabet rather than a representation of the matrix ring.

4. Verified literature review

The following scope statements are supported by publisher metadata/abstracts, proceedings listings, or the cited preprint records; this is not an exhaustive systematic review.

- **Direct matrix-ring cyclic-code work.** Patel, Prakash and Islam study cyclic codes, duals and F_{16}-images over $M_4(F_2+uF_2)$ (the k=2 matrix-ring case); a correction was published. Joshi, Patel and Prakash report the k=3 matrix-ring cyclic-code case at ISIT 2025, and their 2026 arXiv preprint treats cyclic codes over a class of matrix rings for general k. These are direct reasons not to claim new matrix-ring cyclic-code foundations. They do not make the distinct commutative ring R_k isomorphic to the matrix ring. [R16–R18]
- **Matrix-ring quantum adjacency.** A 2026 Quantum Information Processing paper constructs quantum codes from cyclic codes over M_3-matrix rings using a specially chosen inner product and Gray map. It is adjacent methodology, not a direct result for R_k or an M_4 duplicate. [R19]
- **Chain-ring quantum codes.** Biswas and Bhaintwal construct quantum codes from Hermitian self-orthogonal cyclic codes over $F_{(q^2)}[u]/\langle u^k\rangle$, with a Gray map and dual results. This is close prior art for the single-nilpotent chain-ring specialization and means the revised project must not claim that general ring-to-quantum pipeline as new. [R9]
- **Constacyclic quantum codes over non-chain rings.** Published work includes a 2021 non-chain-ring construction, an $A_{(q,r)}$ constacyclic/skew-constacyclic study with Hermitian dual-containment criteria, later constacyclic hull and Gray-map constructions, and a September 2026 paper over another finite non-chain family. Bhardwaj, Goyal and Raka also study constacyclic codes and quantum codes over a general ring whose defining polynomials split into distinct linear factors. These establish the broad method and make any generic “non-chain constacyclic codes yield quantum codes” claim untenable. [R6–R8, R10–R12, R14–R15, R20]
- **Local Frobenius non-chain code structure.** Prior work treats constacyclic codes and duals for certain local Frobenius non-chain rings with small nilpotency index. This is relevant to the ring-theoretic methods, but the exact two-variable ring and the proposed duality-adapted map still require a direct comparison. [R4–R5]
- **Quantum construction.** The Hermitian construction from a dual-containing $F_{(q^2)}$-linear code is classical. Here q=4, so the field-level construction is 4-ary once an actual Hermitian dual-containing image has been established. [R1]

The verified bibliography and source-access limits are recorded in `VERIFIED_REFERENCE_LIST.md`. A publication-level novelty review remains required before claiming priority.

5. Precisely formulated research gap

The broad subjects—constacyclic codes, finite non-chain rings, Hermitian dual containment, Gray maps, and quantum constructions—are established. The project must not claim novelty for combining these keywords.

The **candidate, narrowly stated gap** is:

For the commutative local complete intersection $R_k = F_{16}[U,V]/\langle U^k, V^4\rangle$, odd lengths n, and central multipliers $\lambda \in \mu_5(F_{16})$, determine the CRT component duality and give a necessary-and-sufficient Hermitian dual-containment test for componentwise monomial ideals; then transport that exact duality through a specified F_{16}-linear isometry and identify quantum parameters for verified examples.

A targeted exact-ring search did not surface this theorem package. That observation is not proof of novelty. The gap is accepted only if (i) the component duality is proved, (ii) the search over databases, citations and recent preprints finds no equivalent result, and (iii) resulting code parameters are compared with current best-known data. If an equivalent classification exists, the theorem must be sharpened or the project narrowed before computation.

6. Central research problem

For $k \geq 2$, odd n, and $\lambda^5=1$ in F_{16}, characterize the Hermitian dual-containing λ-constacyclic codes over R_k within the componentwise monomial-ideal family, and determine when their duality-adapted F_{16}-images yield nontrivial 4-ary quantum stabilizer codes.

The all-ideal classification problem is explicitly **not** assumed solved by the monomial analysis.

7. Mathematical preliminaries

- F_{16}/F_4 is a quadratic extension with Hermitian involution $a \mapsto a^4$.
- $F_{16}^{\times}$ has exactly five fifth roots of unity, including 1; the other four have order 5.
- The ring involution on R_k applies $a \mapsto a^4$ to coefficients and fixes U,V.
- A finite commutative Frobenius ring supports the usual cardinality duality for linear codes. The explicit field-valued form needed for the Gray-type map is defined in Section 16 below.
- For a field-linear code $D \subseteq F_{16}^N$, $D^{\perp_H}$ uses the standard Hermitian product $\sum_j x_j y_j^4$. The quantum distance is not automatically the classical minimum distance; it is the minimum weight in the nontrivial logical cosets.

8. Corrected algebraic framework

Set

$R_k = F_{16}[U,V]/\langle U^k, V^4 \rangle$, $k \geq 2$.

Every element has a unique expansion $\sum_{0 \leq a<k,\ 0 \leq b<4} c_{a,b} U^a V^b$. Thus $\dim_{F_{16}} R_k = 4k$ and $|R_k| = 16^{4k}$. The ring is commutative and local with maximal ideal $m=(U,V)$, residue field F_{16}, nilpotency index $k+3$, and one-dimensional socle $F_{16} U^{k-1} V^3$. It is an Artinian complete intersection, hence Frobenius.

This ring is **not** $M_4(F_2[u]/\langle u^k \rangle)$. The latter is noncommutative and has radical quotient $M_4(F_2)$; the revised ring is commutative with residue field F_{16}. Their equal cardinality is only a cardinality match.

9. Definitions and assumptions

Unless a theorem explicitly states otherwise:

1. $k \geq 2$, $n \geq 1$ is odd, and $\lambda \in F_{16}^{\times}$ satisfies $\lambda^5 = 1$.
2. Codewords are vectors in R_k^n, and codes are R_k-submodules. The ring is commutative, so no left/right ambiguity remains.
3. The shift is $T_\lambda(c_0,\ldots,c_{n-1}) = (\lambda c_{n-1}, c_0, \ldots, c_{n-2})$.
4. A code is λ-constacyclic iff it is invariant under T_λ, equivalently an ideal of $R_k[X]/\langle X^n - \lambda \rangle$.
5. The ring-valued Hermitian form is $\langle x,y \rangle_R = \sum_i x_i \mathrm{conj}(y_i)$, and its dual is the set of vectors orthogonal to every codeword.
6. “Monomial component code” means each CRT component ideal is generated by monomials in U,V over its component field. This is a specified subfamily, not a synonym for every ideal.

10. Research objectives

- Prove the ring, involution, and Frobenius-functional facts in a self-contained preliminary section.
- Establish the square-free factorization and CRT components for the stated (n,λ) range.
- Identify the conjugate-reciprocal pairing of CRT factors and derive the dual of a component ideal.
- Give a finite staircase/monomial description of the componentwise monomial family, its cardinalities, annihilators and dual-containment test.
- Construct and verify a duality-adapted F_{16}-linear coordinate isometry, with no unsupported claim of standard Lee-weight preservation.
- After proof gates and researcher approval, build selected examples and verify quantum parameters against exact distances or certified bounds.

11. Research questions

1. How does $\alpha \mapsto \mathrm{conj}\alpha^{-1}$ permute the irreducible factors of $X^n - \lambda$, and what is the induced component-ring isomorphism?
2. What is the exact annihilator staircase of a monomial ideal in $K[U,V]/\langle U^k, V^4 \rangle$?
3. Which component staircases satisfy Hermitian dual containment under the factor-orbit pairing?
4. Can the resulting family contain codes with $K > 2kn$, giving positive quantum dimension $2K - 4kn$, and useful distances?
5. Which verified parameter tuples, if any, are not dominated by current 4-ary quantum-code records?

12. Proposed mathematical constructions

12.1 Constacyclic quotient and CRT

For the stated λ, $X^n-\lambda$ is separable over F_{16}: its derivative is X^{n-1} in characteristic two, and $\gcd(X^n-\lambda, X^{n-1})=1$. Write $X^n-\lambda=\prod_{i=1}^{r} f_i(X)$ as a product of distinct monic irreducibles over F_{16}. The same factorization holds in $R_k[X]$, and the factors remain pairwise comaximal because a Bézout identity over F_{16} remains valid after coefficient extension. Hence

$R_k[X]/\langle X^n-\lambda\rangle \cong \prod_i S_i$,
$S_i \cong K_i[U,V]/\langle U^k, V^4\rangle$,
$K_i = F_{16}[X]/\langle f_i\rangle$.

No Hensel-lifting assertion is needed in this exact setup: the polynomial and its factors are already over the coefficient field. Hensel lifting is not to be invoked for a different coefficient perturbation without checking its hypotheses.

12.2 Monomial component family

Choose in each S_i a monomial ideal I_i. Its support $E_i \subseteq \{0,\dots,k-1\}\times\{0,1,2,3\}$ is upward closed in the truncated exponent rectangle. The CRT product $C \leftrightarrow \prod_i I_i$ is the research family. If $m_i=|E_i|$ and $d_i=[K_i: F_{16}]$, then $|I_i|=|K_i|^{m_i}$, and the intended dimension formula is $K=\sum_i d_i m_i$. This is a monomial-family formula, not a classification of arbitrary component ideals.

12.3 Conjugate-reciprocal component pairing

For a root α of $X^n-\lambda$, the map $\tau(\alpha)=\text{conj}\alpha^{-1}$ is an involution on the root set because $\bar{\lambda}^{-1}=\lambda$. It induces an involution on irreducible factors. The project will define the corresponding coefficient-semilinear component maps $\theta_i: S_i \to S_{\tau(i)}$, fixing U,V, and prove the exact component dual relation, including any harmless unit factor.

12.4 Quantum image

For $a=\sum c_{r,s} U^r V^s$, let $\kappa(a)=(c_{r,s})_{r,s}$. Define $\eta(a)=c_{k-1,3}$. On R_k^n, use $B(x,y)=\eta(\sum_i x_i \overline{y_i})$. The monomial basis has a nondegenerate Hermitian Gram matrix pairing (r,s) with $(k-1-r,3-s)$. Choose an F_{16}-linear isometry T_k from this form to the standard Hermitian form and set $\Phi(x_0,\dots,x_{n-1})=(T_k\kappa(x_0),\dots,T_k\kappa(x_{n-1}))$. The isometry is to be constructed explicitly (for example by Hermitian Gram reduction), not guessed from cardinality.

13. Expected lemmas and theorems

The full assumptions, conclusions, significance and proof obligations are tabulated in `EXPECTED_THEOREMS_AND_PROOF_ROADMAP.md`. The main proposed result is restricted to the componentwise monomial family. No expected theorem is represented as proved.

14. Proposed proof strategies

- Use the monomial basis and socle to verify locality, nilpotency, and Frobenius nondegeneracy.
- Use the derivative and Bézout identities over F_{16} to establish the CRT decomposition.
- Derive the dual component permutation directly from the coordinate Hermitian product / conjugate-reciprocal polynomial operation; do not infer it from cardinalities alone.
- For a monomial ideal, use the truncated monomial basis to derive its annihilator support and cardinality. Prove both inclusions for the annihilator formula.
- Prove the top-socle functional pairing is nondegenerate. Because nondegenerate Hermitian forms of a fixed dimension over F_{16}/ F_4 are equivalent, construct T_k and verify the Gram identity.
- Apply the field Hermitian construction only after dual transfer and dual containment are proved for the image.

15. Duality and quantum-code methodology

The involution on R_k is conj{$\sum$ c_(r,s)U^rV^s}=$\sum$ c_(r,s)^4U^rV^s. Since $\lambda^5=1$, $\bar{\lambda}^{-1}=\lambda$. The proposed dual theorem must show that the Hermitian dual remains in the same λ-constacyclic family and that its CRT exponents/staircases are paired by τ.

For any R_k-linear code C, the intended field-valued pairing B has the same orthogonality relation as the R_k-valued form: if B(c,y)=0 for all c∈ C, then applying this to ac∈ C for every a∈ R_k and using nondegeneracy of (a,z) ↦ η(az) forces ⟨ c,y⟩_(R)=0. Thus the planned isometry should give Φ(C^(⊥_H))=Φ(C)^{⊥_H}. This is a proof target, not a claim about the previous matrix-ring paper's Gray map.

If D=Φ(C) is Hermitian dual-containing, the standard Hermitian construction over F_{16}= F_(4^2) gives a 4-ary stabilizer code with length N=4kn, quantum dimension K_Q=2K-N, and distance

d_Q=min{wt_H(z):z∈ D\ D^(⊥_H)}.

The condition K>N/2 is required for positive quantum dimension. Neither equality of distance with d_H(D) nor purity is assumed.

16. Gray-map framework

The map Φ is a blockwise F_{16}-linear isometry for the explicitly defined Hermitian forms. It is bijective because it is an invertible coordinate transform. It maps R_k-linear codes to F_{16}-linear codes and preserves the exact dual. Since T_k commutes with multiplication by λ∈ F_{16}, the image remains invariant under a block λ-constacyclic shift, hence is quasi-twisted of block index 4k.

No claim is made that coefficient expansion or Φ preserves a separately standard Lee or homogeneous weight. The Hamming metric of the output is the metric used for quantum parameters; if a ring symbol weight is desired, it must be defined as the pullback of Hamming weight or independently proved compatible.

17. Computational methodology

No code search is authorized or performed in this blueprint-finalization stage. After approval and after the algebraic/duality theorem gates pass:

1. Enumerate staircase supports for the stated monomial family, not all ideals of S_i.
2. Compute component duals using the proved annihilator/factor-pairing rule and discard non-dual-containing tuples.
3. Build generator/parity-check descriptions and verify their dimensions independently by CRT cardinalities and field ranks.
4. Construct T_k exactly over F_{16}; check its Gram identity and the image dual relation symbolically.
5. For modest lengths only, calculate exact minimum logical-coset distance by exhaustive enumeration; for larger lengths use a stated certified lower bound. Record reproducible seeds, generators, and software versions.
6. Compare only after code construction, using an explicitly dated best-known-code snapshot, matching field, exact/lower-bound convention, and dominance rule.

18. Expected code parameters and verification strategy

The ambient length is N=4kn. For a monomial component tuple, the expected field dimension is K=$\sum_i$ d_i m_i, where d_i=deg f_i and m_i counts monomials in component I_i. If the expected dual criterion holds, the predicted quantum dimension is 2K-4kn. The distance is not predicted numerically here; it is the minimum nontrivial logical-coset weight and must be exactly computed or rigorously bounded for each reported example.

This blueprint contains no parameter catalogue, no claimed new code tuple, and no assertion that nontrivial λ yields new lengths. For fixed k,n, the field-image length is independent of λ.

19. Expected novel contributions

Subject to the prior-art and proof gates, the candidate contributions are:

1. A ring-specific factor-orbit and monomial-annihilator criterion for Hermitian dual containment over $F_{16}[U,V]/\langle U^k, V^4\rangle$ at odd lengths and fifth-root multipliers.
2. A completely specified top-socle Hermitian pairing and an explicit, verifiable F_{16}-linear isometry that transfers the exact dual for this alphabet.
3. A reproducible construction pipeline from admissible exponent staircases to 4-ary quantum codes, with only certified distances and carefully qualified best-known comparisons.

The Gray/quantum pipeline alone is not novel; similar methods already exist. The exact ring-specific dual classification and any demonstrably new parameters must carry the contribution.

20. Potential limitations and mathematical risks

- The exact ring may already be covered in a paper or thesis not found by targeted searches.
- The component duality may require a nontrivial unit twist; the precise θ_i action and inclusion orientation must be proved.
- Monomial ideals form a restricted subclass; results cannot be described as a classification of all constacyclic codes unless nonmonomial ideals are handled separately.
- The duality-adapted linear map may change Hamming weights relative to a conventional ring metric. Distances must be computed on the actual image.
- Nontrivial quantum dimension and useful distance are not guaranteed by the existence of dual-containing ideals.
- General lengths and k may produce a large staircase/parameter search; computation is deferred until theoretical gates pass.
- Q1 potential cannot be assessed by journal quartiles before a theorem and novelty case exist.

21. Research work plan

1. **Approval and scope lock:** approve the change from matrix alphabet to R_k; agree on k, odd n, and μ_5 scope.
2. **Algebra verification:** prove ring/Frobenius facts, constacyclic ideal correspondence, square-free factorization and CRT.
3. **Duality theorem:** fix reciprocal-conjugate conventions; derive component pairing; prove monomial annihilator and dual-containment criterion.
4. **Field-image theorem:** construct T_k; prove form isometry, exact dual transfer, and block-shift compatibility.
5. **Novelty gate:** search exact ring variants, theses, preprints and citation chains; revise or stop if equivalent work is found.
6. **Approved pilot computation:** enumerate a deliberately small staircase set, verify ranks and exact distances, then compare to a dated database snapshot.
7. **Contribution decision:** retain only theorems and examples that survive proof, independent checking, and novelty review.

This is a plan only; actual research begins only after explicit approval.

22. Complete manuscript outline

1. Introduction and precisely bounded contribution.
2. Ring R_k, Frobenius structure, involution and forms.
3. Constacyclic quotient, square-free factorization and CRT.
4. Monomial component ideals, cardinalities and annihilators.
5. Hermitian dual, conjugate-reciprocal component orbits and dual-containing criterion.
6. Duality-adapted field map and block constacyclic structure.
7. 4-ary quantum construction, exact distance method and certified examples.
8. Comparison with existing quantum records and limitations.

9 Conclusion; full reference list and reproducibility appendix.

23. References

See `VERIFIED_REFERENCE_LIST.md` for the checked bibliography, source links, access status, and exact limits on the novelty claims. The targeted search is a starting point, not a systematic-review certificate.